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Urban feral cats (*Felis catus* L.): perspectives for a demographic control respecting the psycho-biological welfare of the species

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Summary. - The presence of feral cats in the urban environment is creating problems of acceptance by human beings especially in relation to health aspects. In accordance with the protectionist spirit of the law 281 of 14/8/91, the new sensitivity to the general welfare of animals, and better knowledge of their natural history, some possible interventions of the Public Veterinary Services for feral cat control in the cities are proposed. These consist mainly in demographic control by means of neutering the cats, with particular attention to the psycho-biological welfare of the species.

Key words: feral domestic cats, demographic control, welfare.

Riassunto *(Gatti senza padrone nell' ambiente urbano* (Felis catus *L.*): *prospettive per un controllo demografico in armonia con il benessere psico-fisico della specie).* - La presenza dei gatti domestici liberi nell' ambiente urbano sta creando problemi di accettazione da parte dei cittadini in relazione a problemi igienico-sanitari. In ragione dello spirito protezionistico della legge 281 del 14/8/91, della nuova sensibilità per il benessere degli animali, e la migliore conoscenza della loro biologia, vengono proposti degli interventi da parte del Servizio Veterinario Pubblico per il controllo dei gatti liberi nelle città. Questi consistono principalmente nel controllo demografico per mezzo della sterilizzazione dei gatti, con particolare riguardo del benessere psico-fisico della specie.

Parole chiave: gatti urbani, controllo demografico, benessere.

Introduction

The presence of feral cats in the urban environment in Italy has a long history. There are several documents of ancient Rome in which there are descriptions of domestic cats living around monuments, in public gardens and in places where they could find space. In those times it was easier than nowadays. There are no data to judge if cat density was higher or lower compared to the present day, but it is certain that much attention is now being paid to the presence of cats in cities in relationship to social, sanitary and economic problems. Due to their high density, the highest compared to that of other environments where domestic cats live (Table 1) [1], several interventions to attempt to control their populations have been undertaken.

In the past, these interventions were without any kind of control and were carried out by both public administration and private people. They had one common characteristic: cruelty towards the animals. In fact, the only kind of action to "control" the population of domestic cats consisted of killing the animals by means of poison or by shooting them [2]. One episode, in particular, was sadly notorious in Rome: in 1975, several dozens of cats living in the public garden situated in Vittorio Emanuele Square were attracted and trapped in a storehouse that was subsequently burned down. Not one cat survived! And it is well known, although nobody has ever confirmed it officially, that in Largo Argentina (Rome) cats have been periodically killed because "there were too many".

Today, in the light of the protectionist spirit of the law 281 of 14/8/91, possible interventions of the Public Veterinary Services for feral cat control in the cities are proposed. Before examining them, it is worthwhile defining "urban feral cats", and describing the living conditions of feral cats in the cities; furthermore, it is important to consider why their density is higher than in any other environment, and to examine the positive and negative aspects of cat presence in the urban environment.

Definition of urban feral cats

This paper concerns only the demographic control of urban feral cats which have to be distinguished from other categories of domestic cats. The following classification [3] can be useful to identify the subject of the study.

Table 1. - Summary of population density and group sizes for cats studied in different habitats *(*)*

Site	Population density/km	Group size
S.E. Australia	0.74-2.4	1
Orongorongo Valley, N.Z.	2	
Galapagos I.	2.2-2.5	adult groups rare
L'ile aux Cochons	2.5-10	
Swedish farmland	2.5-3.3 (>20)	2-7 females
Heisker (Hebrides, UK)	3.7	
Marion Is. (inland)	4.9	solitary
Marion Is. (coastal strip)	13.8	solitary
Macquarie Is.	4-7	
English farmland	4.3-6.3	1-5 (mode 2, max 24)
Church Farm (Devon)	6.0	4-18
Rural Illinois	6.3	5.6
Kerguelen Is.	6-13	
Rural California	12.5	
W. Scotland	19.0	summer: solitary; winter: groups
Dassen Is.	16.5-22.8	adults solitary; sub-adults 2-6
Portsmouth dockyard	> 200	2-9 females
English industrial		7% >50 47% < 10
Urban Rome	> 2000	25-34
Urban Baltimore	740	
Ainoshima fish market	2350	max 19

(*) Reproduced with kind permission from [1].

Domestic cats

Synanthropic cats. - Domestic cats *(Felis catus)*, that live in the same environment as human beings. To this category belong:

a) *cats living exclusively at home*: these never have contact with the world outside the home, they are completely dependent on food provided by human beings and do not hunt. Thus, they can be considered "dependent on human beings" in terms of trophic resources.

b) *urban feral cats*: domestic cats that live freely in the urban environment. They are dependent on food provided either by human beings (cat-lovers) or found in refuse; they hunt very rarely. Thus, they can also be considered "dependent on human beings" with respect to trophic resources.

c) *category intermediate between (a) and (b)*: domestic cats that have an owner but which live both inside and outside the home. Thus, they have contact with domestic cats belonging to category (b). They are dependent on food provided by the owner, but they can have access also either to food provided by cat-lovers in the street or to remnants found in refuse. Thus, they can also be considered "dependent on human beings" as far as trophic resources are concerned.

Rural cats. - Domestic cats *(Felis catus)* that live freely around a farm in the rural environment. They are dependent in part on food provided by the owner, but also on hunted prey. Thus, they can be considered "semi-dependent on human beings" in regard to trophic resources.

Spatial and social organization

In the urban environment feral cats live in groups [4, 5]. There is ample evidence that the groups are not simple aggregations lacking either social structure or adaptive significance, but that they can be considered social groups [1, 2, 6]. It has been reported that within the boundaries of the territory of a group they show an active spacing pattern related to the resources (food and shelter) offered by the environment [2]; that there is a linear dominance hierarchy among adult males [5, 7]; and that female cats cooperate in rearing offspring [7]. All these characteristics are typical of species that form social groups.

Resources dispersion hypothesis

It is interesting to note that cats vary greatly in spatial organization, from being solitary in well-dispersed populations at a low density of around one cat per square

km or less [8], to living in groups in clumped populations at high densities of up to 2350 cats per km [2, 4] (see Table 1). There is a continuum between these two extremes (reviewed by [9]).

The main hypothesis to explain these differences in spacing pattern is the "resources dispersion hypothesis" [10]. This theory suggests that where the food resources are clumped and abundant, the benefits of sociality are greater than costs; where food resources are scant and well-dispersed, the costs of sociality overcome the benefits. Since in the urban environment there is a great amount of food available for cats (the highest in any environment) provided both by cat-lovers and by remnants found in rubbish, the animals show the highest density of any environment, and gather in social groups.

Positive and negative aspects of cat presence in the urban environment

Among the positive aspects of the presence of cats in the urban environment [3], the following have been identified: a) predatory; b) anti-depressive; c) educative; and d) aesthetic effects. Negative aspects concern mainly: a) hygienic considerations and b) public health problems.

Positive aspects:

a) *Predatory action*: surprisingly, given the importance of the phenomenon, the predatory impact of urban feline populations on rodents has never been evaluated in Italy. Although it seems that even an adult cat is not always able to prey on an adult rat (*Rattus norvegicus* or *R. rattus*), it is possible that domestic cats can nevertheless exert a controlling action on rodent demography, at least on mice populations. In fact, Cristaldi *et al.* have carried out some research on rodent populations in Rome and reported that it was more difficult to trap mice (*Mus domesticus*) where there was a conspicuous presence of domestic cats, probably because the density of mice was lower than in other places where there were no cats [11]. Some years later in the same place, there were no cats and it was much easier to trap mice (Cristaldi, personal communication). It would be extremely interesting to check if and how the feral cats exert control on the demography of rodents in Rome.

Moreover, it is well known, but poorly considered by scientific behavioural studies, that a strong deterrent for rodent presence in the house is the presence of a domestic cat.

b) *Anti-depressive effects*: there is a very important role played by feral cats in cities, which consists mainly of an anti-depressive effect on the people that look after them. In many Italian cities there is a strong tradition of taking care of feral cats, and these cat-lovers are often elderly and lonely people who find a purpose in life when looking after these animals.

c) *Educative effects*: in an increasingly urbanized world, the presence of social groups of domestic cats is a unique resource for learning something about the natural world and about the behaviour of animals. This is true not only for children, but also for adults.

d) *Aesthetic effects*: it is important not to forget that domestic cats are considered pleasant from an aesthetic point of view by the majority of people, even by those who dislike animals in general. In Italy, as I said before, there is a strong tradition to look after feral cats because they have been present in the urban environment for many centuries. They are a sort of "living decoration" that has become inseparable from the image of some cities (e.g. Rome and Venice).

Negative aspects:

a) *Hygienic problems*: the "environmental impact" of cat presence takes the form of deposits dispersed in the environment (faeces and sprays of urine), as well as rubbish (sheets of paper, containers for food) that remain on the ground. These, together with caterwauling, can cause reactions of intolerance from human beings.

b) *Public health problems*: another "cost" of cat presence to take into account is that the domestic felines can be sources of zoonoses, i.e. illnesses transferable to human beings (for details see [3]).

Suggestions for demographic control of feral cats in cities

On the basis of the considerations made above and of law 281 of 14/8/91 it is possible to affirm that urban feral cat populations have *not* to be destroyed but that some kind of demographic control should be developed and applied for the welfare of the animals (see below).

Before undertaking any kind of action, however, it is of crucial importance to know the local situation in detail. This requires an accurate investigation of any colony in which control has to be exerted, i.e. at least on the kind of environment in which the colony lives and on the number of cats belonging to that colony. It would also be useful to have a map indicating cat populations in the city.

Traditionally, in Italy the control of urban feral cats was performed by Public Veterinary Services and was based on the legislation on rabies control. Besides such services, there were in many Italian cities like Venice, Milan, Genoa, Rome, etc. many private initiatives, performed by single persons or associations of cat-lovers. As already said, cat love is deeply rooted in the people of Rome. Furthermore, two recent important facts have influenced the cat/human relationships in Rome, and in Italy: the first is the law 281/91, and the second is the constitution of an Office for Animal Rights established by the Assessorate of Culture of Rome.

Thus in Rome, and also in other Italian cities, the Public Veterinary Services which exert a demographic control on feral cat colonies contact the cat-lovers and/or the voluntary associations that look after the animals

because these know the colonies much better than anyone else, expecially the dynamics of the groups. These people are usually willing to collaborate, can indicate which animal is ill and needs sanitary help and, more importantly, the cats know them and can be trapped by them with less stress than from strangers.

To further protect the psycho-biological welfare of the species (see below) and to save time and energy, it is important to collaborate with an ethologist who can suggest the best intervention strategy.

The next step is a sterilization campaign carried out respecting the welfare of the animals. The Public Veterinary Services (USL), in collaboration with the Zoophilist Associations present in the zone, trap, sterilize and then return the animals to the original colony. Before being returned to their colony, the sterilized cats have to be marked, thus avoiding repeated trapping and anaesthesia of cats which have already been neutered. The best method is to remove the tip of one ear; other methods are less effective [12].

The psycho-biological welfare of the species

It is mentioned above that "some kind of demographic control should be exerted for the welfare of animals themselves". This statement is based on quantitative data. Actually, in urban cat colonies kitten mortality is very high, reaching sometimes 90% of the young born each year [5]. One of the costs of sociality is the ease of the transmission of some diseases, for example of rhinotracheitis. Since female cats belonging to the same social groups tend to cooperate in rearing offspring [7], one of the consequences is that a great number of kittens play, eat and sleep together, facilitating the transmission of disease. Considering that in large groups of cats there can easily be 30 or 40 kittens reared together, and that most of them die miserably before reaching one year of age, it is clear that it is important to limit births.

Another important point is collaboration with an ethologist, who knows in detail the social organization and mating behaviour of the species. In the past, the Public Veterinary Services have chosen the wrong strategy for neutering cats, castrating only the dominant male of the group. This decision was based on the belief that the dominant male of the group monopolizes all copulations. But it has already been shown [5, 6] that female cats copulate with more than one male, and even with males strange to the group. Thus, to reach the goal of an adequate cat population control, it is important not only to sterilize all the males of the colony, but also to neuter the adult females.

Furthermore, it has been shown that the social hierarchy of a group does not change if all animals are sterilized [13], whereas a reorganization of social relationships is necessary if only some individuals are neutered. Since one of major causes of stress in all social species is the reorganization of social relationships, it is clear that in order to keep inevitable stress at a low level it is necessary to intervene on all animals of the group.

It has however been shown that for an adult cat it is practically impossible to be integrated into an already established social group. Especially males, but also females, that were abandoned by human beings as strangers into an already well-established colony of cats, were found dead after some days without any morphological alteration [5]. Observations revealed that abandoned adult cats were chased all day long, were not able to feed, and were clearly stressed. Thus, the most important point to consider in respect of the psycho-biological welfare of cats, is to release them back into the original colony after being neutered. These are just a few examples of the role an ethologist can play in a sterilization campaign, but there are many other cases where such a collaboration would help choose the best intervention strategy.

There has also been a debate about which method of sterilization should be used. Attempts at using contraceptives have been abandoned because of the difficulties of this approach with animals that are not tame [13]. So far, the most accepted method of sterilization is ovariectomy for females and orchiectomy for males because they eliminate, besides fertility, most of the causes of disturbance for human beings, i.e. urine sprays and vocalizations. These surgical operations are also best at limiting the stress for the animals because other interventions that do not eliminate the gonads cause repeated oestrus in females and excessive sexual activity in males. Sterilized animals tend to become more sedentary and still show territorial behaviour, thus defending their area against conspecifics strange to the group [13], occupying thus the space that is not available to other cats.

Conclusions

The most important points that emerge in this paper are that the demographic control of urban feral cats has to be done on the basis of interdisciplinary collaboration, and that the heart of the matter lies in sterilization of the animals. There are many details accompanying the neutering of cats that are in harmony with the psycho-biological welfare of the species. But the most important question concerning the welfare of the animals remains: where are we going? This is a question that all people involved in the management of urban animals should seriously consider. Is a world of sterilized feral cats and fertile cats of valued breeds desirable? Probably not, but it is also probable that we shall never succeed in sterilizing all the feral cats of a city. Nevertheless, although a new

sensitivity to the general welfare of animals, a better knowledge of their natural history, and the protectionist spirit of law 281 of 14/8/91 have improved the situation of feral cat control in the cities, there is still the need for more general plans for interventions that should take into account a perspective for the future.

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